

Distributed Optimization for Machine Learning

Lecture 18 - Data heterogeneity in decentralized optimization

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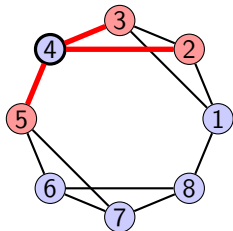
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Last-lecture: Decentralized SGD for distributed learning

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i} [F(\mathbf{x}; \xi_i)].$$



Every node has 3 neighbors.

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- **Idea:** Instead of global averaging, partial averaging with neighbors

Benefit and limitation: Reduce comm. cost to $O(d_{\max})$ but need to design a double-stochastic mixing matrix.



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Partial averaging as one-step average consensus

- Write the DSGD in the matrix form.

DSGD

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^{(k)}; \xi_i^{(k)})$$

$$\mathbf{x}^{k+1} = W \mathbf{x}^{k+\frac{1}{2}}$$

where $\mathbf{x}^l = [\mathbf{x}_1^l, \dots, \mathbf{x}_n^l]$, $l = k, k + \frac{1}{2}$.

- Recall average consensus algorithm (fixed weight matrix)

$$\mathbf{x}(t+1) = W \mathbf{x}(t) = W^t \mathbf{x}(0)$$

- The partial averaging in DSGD is one-step ($t = 1$) version of average consensus algorithm with varying $\mathbf{x}(0) = \mathbf{x}^{k+\frac{1}{2}}$



Convergence rate of DSGD

Assumptions:

- f_i : L -smooth
- $\nabla F(\mathbf{x}; \xi_i)$ is an **unbiased estimate** of $\nabla f_i(\mathbf{x})$, with **bounded variance**
- Bounded data heterogeneity: $\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x) - \nabla f(x)\|^2 \leq b^2$
- The weight matrix **W** is doubly stochastic

Theorem 1 (Convergence rate of DSGD)

Suppose above assumptions hold and let $\eta = 1/\sqrt{K}$. Then

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\bar{\mathbf{x}}^k)\|_2^2] \leq \mathcal{O} \left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3} \sigma^{2/3}}{K^{2/3}(1-\rho)^{1/3}} + \frac{\rho^{2/3} b^{2/3}}{K^{2/3}(1-\rho)^{2/3}} \right)$$



Recall linear speedup in parallel SGD

Theorem 1 (Parallel SGD under smooth loss)

Let f be L -smooth and $\eta_k \equiv \eta = 1/\sqrt{K}$. Then the following holds

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\mathbf{x}^k)\|_2^2] \leq \mathcal{O}\left(\frac{1}{K} + \frac{\sigma}{\sqrt{nK}}\right)$$

This implies that to achieve an ϵ -accurate solution, Parallel SGD needs

$$\frac{\sigma}{\sqrt{nK}} \leq \epsilon \quad \implies \quad K \geq \frac{\sigma^2}{n\epsilon^2} \quad \text{iterations,}$$

which decreases linearly with number of nodes n ; called *linear speedup*!



Linear speedup in DSGD

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\bar{\mathbf{x}}^k)\|_2^2] \leq \mathcal{O} \left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3} \sigma^{2/3}}{K^{2/3} (1 - \rho)^{1/3}} + \frac{\rho^{2/3} b^{2/3}}{K^{2/3} (1 - \rho)^{2/3}} \right)$$

- The first term dominates when $K \rightarrow \infty$, which suggests DSGD achieves *linear speedup* asymptotically.
- Convergence of DSGD depends on the *network topology*: sparse topology ($\rho \rightarrow 1$) results in slower convergence.
- Convergence of DSGD depends on the *data heterogeneity*: large heterogeneity b results in slower convergence.

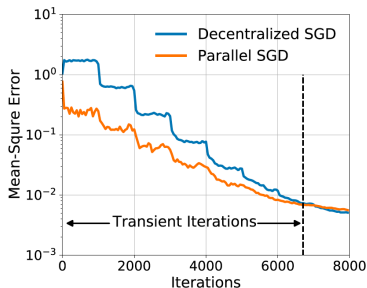


PSGD v.s. DSGD

PSGD: $\mathcal{O}\left(\frac{\sigma}{\sqrt{nK}}\right)$ **Transient iterations:** Extra overhead $\leq \frac{\sigma}{\sqrt{nK}}$

Extra overhead

DSGD: $\mathcal{O}\left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3}\sigma^{2/3}}{K^{2/3}(1-\rho)^{1/3}} + \frac{\rho^{2/3}b^{2/3}}{K^{2/3}(1-\rho)^{2/3}}\right)$



- DSGD can asymptotically converge as fast as P-SGD

- Transient iterations:

$$\mathcal{O}\left(\frac{\rho^4 n^3}{\sigma^2(1-\rho)^2} + \frac{\rho^4 n^3 b^4}{\sigma^6(1-\rho)^4}\right)$$

affected by *network topology* and *data heterogeneity*.



Why DSGD suffers from data heterogeneity?

DSGD

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^{(k)}; \xi_i^{(k)})$$

$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- Without gradient noise, decentralized gradient descent follows

$$(\mathbf{DGD}) \quad \mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{x}_i^k - \eta \nabla f_i(\mathbf{x}_i^k))$$

- When achieving stationary: $\mathbf{x}_i^k = \mathbf{x}^*$ for all i . Then the next step is

$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{x}^* - \eta \nabla f_i(\mathbf{x}^*)) = \mathbf{x}^* - \eta \sum_{j \in \mathcal{N}_i} w_{ij} \nabla f_i(\mathbf{x}^*)$$



Why DGD suffers from data heterogeneity?

- In homogeneous scenario: $\nabla f_i(\mathbf{x}^*) = 0$ for all i . Then the stationary point $\mathbf{x}^*$ is stable because

$$\mathbf{x}_i^{k+1} = \mathbf{x}^* - \eta \sum_{j \in \mathcal{N}_i} w_{ij} \nabla f_i(\mathbf{x}^*) = \mathbf{x}^*$$

- In heterogeneous setting: $\nabla f_i(\mathbf{x}^*) \neq \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}^*) = \nabla f(\mathbf{x}^*) = 0$.

$$\mathbf{x}_i^{k+1} = \mathbf{x}^* - \eta \sum_{j \in \mathcal{N}_i} w_{ij} \nabla f_i(\mathbf{x}^*) \neq \mathbf{x}^*$$

suggesting the stationary point is not stable.

- Cause divergence when $b^2 \triangleq \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2$ is large.

Q: How to alleviate this issue?



Inexact convergence of DGD with constant stepsize

Assumptions:

- Each f_i is L -smooth and μ -strongly convex
- $\mathbf{W}$ is doubly stochastic, and the network is connected

Theorem 2 (Inexact convergence with constant stepsize)

Let DGD iterate as

$$\mathbf{x}^{k+1} = \mathbf{W}\mathbf{x}^k - \eta \nabla f(\mathbf{x}^k),$$

where $\eta \in (0, \frac{1+\rho}{L})$ and $\rho = \max_{i \geq 2} |\lambda_i(\mathbf{W})|$. Then:

$$\|\mathbf{x}^k - \bar{\mathbf{x}}^k\| = \mathcal{O}(\eta), \quad \limsup_{k \rightarrow \infty} \|\mathbf{x}^k - \mathbf{x}^*\| = \mathcal{O}\left(\frac{\eta}{1-\rho}\right).$$

Interpretation: With a constant stepsize, DGD converges only to a *neighborhood* of the global optimum, whose size shrinks linearly with η and depends on the network connectivity $(1 - \rho)$.



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Inexact convergence of decentralized SGD

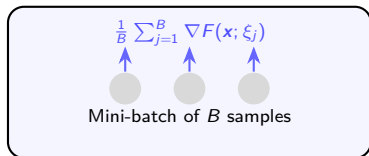
Gradient tracking: tackling data heterogeneity

Byzantine robustness in decentralized networks



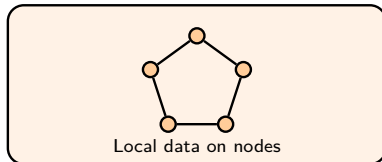
Insights from mini-batch SGD

Single machine



Mini-batch SGD

Decentralized network



Decentralized SGD

- Gradient $\nabla f_i(\mathbf{x})$ for each node can be viewed as a "mini-batch" gradient with respect to the the global gradient $\nabla f(\mathbf{x})$.
- Data heterogeneity $b^2 \triangleq \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2$ is in the same spirit of "variance", but is for each worker.

A: Applying 'variance reduction' technique to local workers.



Recall: how SVRG reduces the variance

Key idea: SVRG replaces noisy gradients with a corrected version that re-centers them around the full gradient at a *snapshot point*.

- Periodically calculate the **full gradient** at a "snapshot" point $\tilde{\mathbf{x}}$.
- Use this full gradient as a "low-variance anchor" to correct

$$\mathbf{v}_{\text{SVRG}}^t = \nabla f_{i_t}(\mathbf{x}^t) - \nabla f_{i_t}(\tilde{\mathbf{x}}) + \nabla f(\tilde{\mathbf{x}})$$

- f_{i_t} : gradient for the current random sample i_t .
- $\nabla f(\tilde{\mathbf{x}})$: full gradient at a snapshot point (updated every epoch) $\tilde{\mathbf{x}}$.

How to track the global gradient in decentralized setting?



Recall: The SVRG algorithm

for epoch $s = 1, 2, \dots$

- Compute the full gradient at the snapshot:

$$\nabla F(\tilde{\mathbf{x}}) = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{\mathbf{x}}) \quad \text{set snapshot } \tilde{\mathbf{x}} \text{ from the last epoch}$$

- **for** inner iteration $t = 1, \dots, m$

- Choose a sample i_t uniformly at random and compute

$$\mathbf{v}_{\text{SVRG}}^t = \nabla f_{i_t}(\mathbf{x}^t) - \nabla f_{i_t}(\tilde{\mathbf{x}}) + \nabla F(\tilde{\mathbf{x}})$$

- Run the gradient descent update: $\mathbf{x}^{t+1} = \mathbf{x}^t - \eta \mathbf{v}_{\text{SVRG}}^t$

end for

end for

- Full gradient calculation is expensive but only done once per epoch.
- Inside the epoch, computations are two stochastic gradients.



Track the global gradient

Key idea: introduce additional variable y which converges to the global gradient when $\nabla f_i(\mathbf{x}_i^k)$ achieves stationary point.

- Initialize the tracking variable for each node as $\mathbf{y}_i^0 = \nabla f_i(\mathbf{x}_i^{(0)})$.
- Synchronize the smoothed version with neighbors

$$\mathbf{y}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{y}_j^k + \nabla f_j(\mathbf{x}_j^{k+1}) - \nabla f_j(\mathbf{x}_j^k))$$

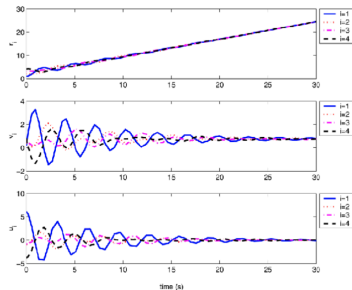
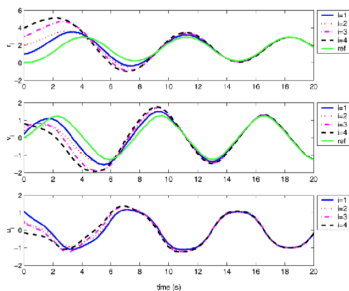
- When each node reaches stationary, the gradient difference of two steps vanishes and

$$\mathbf{y}_i^{k+1} \rightarrow \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}_i^k) \leftarrow \text{global gradient}$$



Tracking variable converges to global gradient

- Green: global gradient



- Tracking variable converges to the global gradient



Simulation results are from [Ren IEEE TAC 2007]

Gradient tracking algorithm

- Consider the general setting where we have gradient noise. Gradient tracking algorithm is defined as follows.

$$\begin{aligned}\mathbf{x}_i^{k+1} &= \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{x}_i^k - \eta \mathbf{y}_i^k) \\ \mathbf{y}_i^{k+1} &= \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{y}_j^k + \nabla F(\mathbf{x}_j^{k+1}; \xi_j^{k+1}) - \nabla F(\mathbf{x}_j^k; \xi_j^k))\end{aligned}$$

- **Key difference:** introduce a tracking variable that can converge to the global averaged gradient asymptotically



Convergence rate of Gradient Tracking

Assumptions:

- f_i : L -smooth
- $\nabla F(\mathbf{x}; \xi_i)$ is an **unbiased estimate** of $\nabla f_i(\mathbf{x})$, with **bounded variance**

Theorem 3 (Convergence rate of GT)

Suppose above assumptions hold and with proper η . Then

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\bar{\mathbf{x}}^k)\|_2^2] \leq \mathcal{O} \left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3} \sigma^{2/3}}{K^{2/3} (1 - \rho)^{1/3}} \right)$$

- **Key feature:** remove the **bounded data heterogeneity** assumption!



DSGD v.s. Gradient Tracking

$$\text{DSGD: } \mathcal{O}\left(\frac{\sigma}{\sqrt{nK}} + \boxed{\frac{\rho^{2/3}\sigma^{2/3}}{K^{2/3}(1-\rho)^{1/3}} + \frac{\rho^{2/3}b^{2/3}}{K^{2/3}(1-\rho)^{2/3}}}\right)$$

Extra overhead

$$\text{GT: } \mathcal{O}\left(\frac{\sigma}{\sqrt{nK}} + \boxed{\frac{\rho^{2/3}\sigma^{2/3}}{K^{2/3}(1-\rho)^{1/3}}}\right)$$

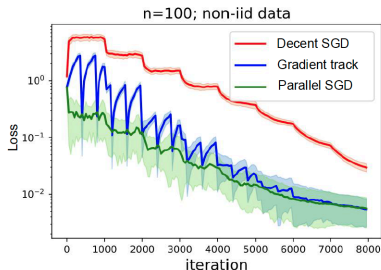
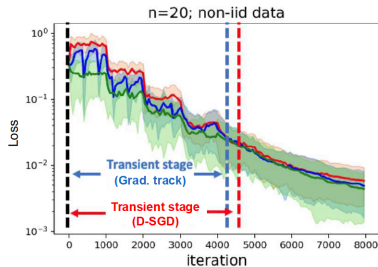
Extra overhead

- Gradient Tracking shorten the transient stage
- DSGD: $\mathcal{O}\left(\frac{\rho^4 n^3}{\sigma^2(1-\rho)^2} + \frac{\rho^4 n^3 b^4}{\sigma^6(1-\rho)^4}\right) \longrightarrow$ GT: $\mathcal{O}\left(\frac{\rho^4 n^3}{\sigma^2(1-\rho)^2}\right)$



Empirical studies on heterogeneous data

- DNN training on ring graph ($1 - \rho = O(n^{-2})$)



- Gradient tracking has shorter transient period.
- Gradient tracking outperforms DSGD on heterogeneous data.



Simulation results are from [Ren IEEE TAC 2007]

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Challenge of robust aggregation under non-i.i.d. data

Design principle so far:

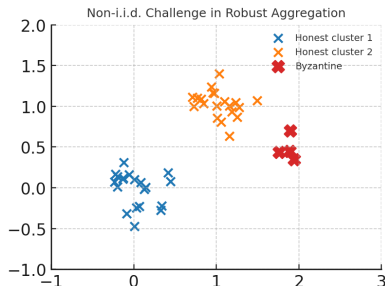
$$\text{Gradient consistency} \quad \mathbf{g}_1 \stackrel{d}{=} \mathbf{g}_2 \stackrel{d}{=} \dots \stackrel{d}{=} \mathbf{g}_n \sim \nabla F(\mathbf{x}; \xi)$$

Challenges:

- Honest nodes may send statistically different gradients.

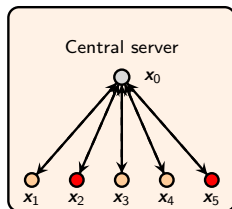
$$\mathbf{g}_1 \stackrel{d}{\neq} \mathbf{g}_2 \stackrel{d}{\neq} \dots \stackrel{d}{\neq} \mathbf{g}_n$$

- Hard to distinguish malicious gradients from natural heterogeneity.



Robust aggregation based on model consistency

Model consistency $\mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n \sim \underset{\mathbf{x}}{\operatorname{argmin}} f(\mathbf{x})$



Consensus-based formulation:

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}_i), \quad \text{s. t. } \mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n.$$

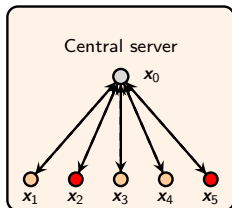
Q: Can we solve this ideal problem if some nodes are Byzantine? **A:** No

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} f_i(\mathbf{x}_i), \quad \text{subject to } \mathbf{x}_1 = \dots = \mathbf{x}_i, \text{ for } i \neq \text{Byzantine}.$$



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

Penalty formulation for aggregation under non-i.i.d. data



Penalty-based formulation:

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} (f_i(\mathbf{x}_i) + \lambda \|\mathbf{x}_i - \mathbf{x}_0\|_1)$$

when $\lambda \geq \lambda_0$ with some threshold λ_0 , it is equivalent to the constrained formulation.

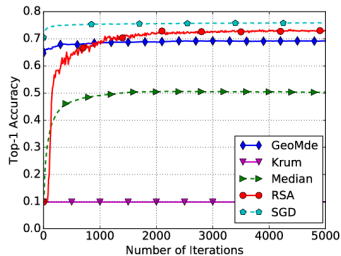
$$\mathbf{x}_i^{k+1} = \mathbf{x}_i^k - \eta (\nabla F(\mathbf{x}_i^k; \xi_i^k) + \lambda \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)) \quad (\text{Update at Honest})$$

$$\mathbf{x}_0^{k+1} = \mathbf{x}_0^k - \eta \lambda \underbrace{\sum_{i=1}^n \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)}_{\substack{\text{include Byzantine} \\ \text{subgradient of } \|\mathbf{x}_i - \mathbf{x}_0\|_1}} \quad (\text{Global aggregation})$$

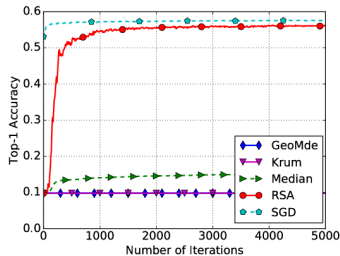


L. Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

Performance of robust aggregation under non-i.i.d. data



(a)



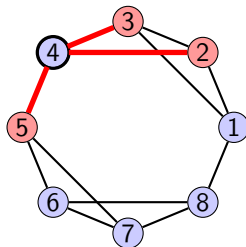
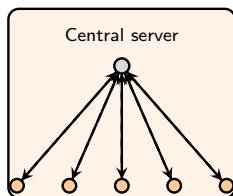
(b)

Figure: Top-1 accuracy under attacks on non-i.i.d. data from MNIST dataset with non-uniform distributed labels: (a) $b = 4$; (b) $b = 8$ out of $n = 20$ nodes.



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

What if we do not have a centralized aggregator?



Motivation:

- Many large-scale systems operate without a trusted central server.
- Examples: allreduce parallel training, federated edge learning.
- Byzantine nodes can exist anywhere in the network.



Vulnerability to Byzantine nodes

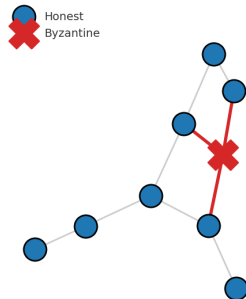
At time k , each node i runs one (**can be multiple**) local SGD steps:

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

When Byzantine nodes exist:

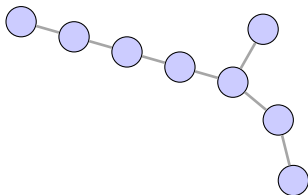
$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}} + (\text{malicious noise})$$

- A single Byzantine can inject large errors.
- Errors may propagate through the network.
- Simple averaging fails catastrophically.

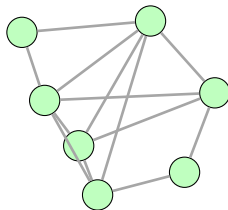


Example: Resilient vs. vulnerable topologies

Sparse Graph (Vulnerable)



Dense Graph (Resilient)



Left: Sparse graph - easy to isolate and corrupt.

Right: Dense, redundant graph - retains consensus even under attack.

Robustness depends on graph structure:

- Honest node must have at least $2b + 1$ neighbors for b Byzantines.
- Honest subgraph must remain connected after removing b nodes.



Robust gossip aggregation

Idea: Replace local averaging with robust aggregation among neighbors

$$\mathbf{x}_i^{t+1} = \text{RobustAgg}(\{\mathbf{x}_j^t : j \in \mathcal{N}_i\})$$

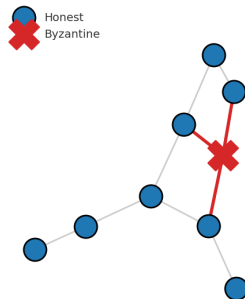
to limit local influence of Byzantine nodes.

Examples:

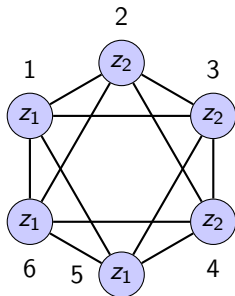
- Coordinate-wise median of neighbors.
- Geometric median over neighbor messages.

Goal: Design robust aggregators $\{\mathcal{A}_i\}$ *through local communications*

- **Consensus** - even under Byzantine nodes.
- **Convergence** - if honest subgraph is well-connected.



Challenges of using median-based robust aggregation



Concrete calculation (next slide):

$$\mathcal{N}_1 = \{2, 3, 5, 6\}, \quad \mathcal{N}_4 = \{5, 6, 2, 3\}$$

For node 1, **the median of** $\{z_1, z_1, z_1, z_2, z_2\}$ is $\boxed{z_1}$; for node 4, **the median of** $\{z_2, z_2, z_2, z_1, z_1\}$ is $\boxed{z_2}$. The two clusters $\{1, 2, 3\}$ and $\{4, 5, 6\}$ remain stuck at z_1 and z_2 .

No consensus is ever reached (even with no Byzantine nodes)!

Z. Wu, T. Chen, Q. Ling. "Byzantine-Resilient Decentralized Stochastic Optimization With Robust Aggregation Rules", *TSP 2023*



Recall convergence of consensus average (Lecture 12)

- Recall the average consensus algorithm (fixed weight matrix)

$$\mathbf{x}(k+1) = \mathbf{W}\mathbf{x}(k) = \mathbf{W}^k \mathbf{x}(0) \quad \text{where} \quad \mathbf{x}(0) = \mathbf{z} \in \mathbb{R}^N$$

Theorem (Convergence rate of consensus)

If $\mathbf{W}$ is doubly stochastic, it holds for the average consensus protocol that

$$\left\| \mathbf{x}(k) - \frac{\mathbf{1}\mathbf{1}^T \mathbf{z}}{N} \right\| \leq \rho^k \|\mathbf{z}\|,$$

where $\rho = \max_{i \geq 2} |\lambda_i(\mathbf{W})| < 1$.

Q: Is the virtual mixing matrix good enough for convergence?

A: The induced $\mathbf{W}$ is **double stochastic**, but its spectrum gap is 0.

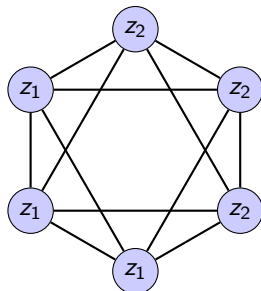


Challenges of designing robust gossip aggregation

Virtual mixing matrix:

$$\bar{\mathbf{x}}_i = \sum_{j \in \mathcal{N}_i \cup \{i\}} w_{ij} \mathbf{x}_j, \quad \mathbf{W} = [w_{ij}]$$

$\mathbf{W}$ induced by robust aggregation is often only **row stochastic**.



Two key challenges:

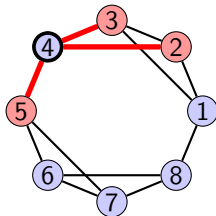
1. **Disagreement:** The *virtual* mixing matrix $\mathbf{W}$ may be row stochastic. Even without Byzantine nodes, some rules (e.g., coordinate median, GeoMed) can cause honest nodes to converge to different points.
2. **Asymmetry:** The *virtual* mixing matrix $\mathbf{W}$ may be non-doubly stochastic, leading to convergence toward $\sum_i p_i \mathbf{z}_i$ instead of $\frac{1}{n} \sum_i \mathbf{z}_i$.



Non-i.i.d. complication in decentralized networks

Challenge:

- Nodes with non-i.i.d. data can appear “suspicious.”
- Natural diversity can mimic Byzantine behavior.



Penalty-based formulation still works

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \in \mathcal{H}} \left(f_i(\mathbf{x}_i) + \lambda \sum_{j \in \mathcal{H}_i} \|\mathbf{x}_i - \mathbf{x}_j\|_1 \right)$$

when $\lambda \geq \lambda_0$ with some threshold λ_0 , it is equivalent to the constrained formulation.

Recent research ideas:

- Cluster-based gossip (different groups exchange robustly).
- Adaptive trust scores per neighbor based on temporal consistency.



Recap and fine-tuning

- What we have talked about **today**?

⇒ **Decentralized SGD** achieves the same convergence rate as Parallel SGD, only to the neighborhood.

⇒ **Gradient Tracking** tackles the data heterogeneity issue by tracking the global averaged gradient.

⇒ **Byzantine-robust gossip aggregation** is challenging to design.



Welcome anonymous survey!

